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ARTICLE

Graphite N-C-P dominated three-dimensional nitrogen and phosphorus co-doped holey graphene foams as high-efficiency electrocatalysts for Zn-air battery

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The search for metal-free catalysts for oxygen reduction reactions (ORR) in energy storage and conversion devices, such as fuel cells and metal-air batteries, is highly desirable but challenging. Here, we have designed and synthesized a controllable 3D nitrogen and phosphorous co-doped holey graphene foams (N,P-HGFs) as a high-efficiency ORR catalyst through structural regulation and electronic engineering. The obtained catalyst shows a half-wave potential of 0.865V in alkaline electrolytes. It is found Zn-air batteries with the N,P-HGFs-1000 air electrode present excellent discharge performance and durability. Our study suggests that the remarkable ORR performance of N, P co-doped graphene is mainly due to the graphite N-C-P structure, where an enhanced charge density and increased HOMO energy level are confirmed by both experimental results and theoretical density-functional theory calculations.

Introduction

Recently, as a new generation of environmentally friendly and efficiently sustainable energy conversion devices, fuel cells and metal-air batteries have attracted considerable attention.¹ However, many challenges still remain.² One of the most significant challenges is the inherently slow ORR kinetics on cathode, which severely limits their development in commercial application.³ In general, Pt-based materials have been regarded as one of the most effective electrocatalysts for the ORR.⁴ However, as scarce, high-cost, low-stability, and methanol-resistance resources, they face a difficult challenge for practical applications.⁵ Therefore, it is highly desirable to study non-precious metal catalysts (NPMCs) as a promising alternative for ORR.^{6,7}

Among various types of NPMCs, metal-free carbon-based materials have been considered as one of the most promising candidates for replacing platinum-based catalysts due to their lower cost, great abundance and high activity.^{8,9} The improved electrocatalytic performance can be attained by tuning the electronic and geometric properties of ORR catalysts. Since the vertical array of N-doped carbon nanotubes (N-CNTs) has been proved to be a highly active ORR catalyst,¹⁰ tremendous efforts have been devoted to optimizing the structures (e.g. 1D¹¹ /

2D^{12,13} / 3D^{14,15} nanostructure) and modulating the doping elements (N,¹⁶ P,¹⁷ S,¹⁸ B,¹⁹ F²⁰) to further improve the electronic conductivity, mass transfer performance, as well as the catalytic activity of the catalyst. As one type of abundant carbon materials, graphene has shown remarkable performance on electrocatalytic applications, mainly due to its superior charge conductivity, large surface area, and abundant doping characteristics.^{21,22} In particular, a great advantage of the unique 3D graphene structure is that it can avoid agglomeration and stacking during pyrolysis, which will influence catalytic active sites of graphene-based materials for ORR.^{23,24}

On the other hand, heteroatom doping is a remarkable way to boost the ORR activity through modulating the charge distribution and spin density.²⁵ Due to differences in the electronegativity and atomic radii of dopant atoms, heteroatomic doping can lead to changes in both structural and electronic properties of the carbon skeleton, providing abundant active sites and defects.²⁶ For instance, nitrogen-doped carbon-based materials have been widely investigated because of their improved electrocatalytic activity for ORR.^{27,28} Nakamura et al. found that the doping of nitrogen atoms in the carbon skeleton facilitates a positive charge of C atoms adjacent to the pyridinic N, and the positively charged carbon atoms can readily bound to O₂ molecules, thus accelerating the ORR process.²⁹ In addition, compared with the single doped carbon-based materials, the binary doped carbon-based materials including nitrogen atoms are expected to exhibit improved ORR performance due to the synergistic effect between the doped heteroatoms.³⁰⁻³⁴ Recently, the phosphorus and nitrogen co-doped carbon-based materials have shown great promise as ORR catalysts,³⁵ while many

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challenges still remain, particularly in verifying active sites during electrochemical processes.³⁶⁻³⁸ Shen and co-authors have found that the excellent ORR performance can be attributed to the existence of a large amount of N-P bonds and pyridinic N in the catalyst.³⁹ Gou et al. proposed that ORR activity is relevant to the concentration of pyridinic-N and quaternary-N, as well as the synergistic effects triggered by P-doping.⁴⁰ The study by Xing et al. suggests that graphitic N and P atom can polarize the adjacent carbon atoms, resulting in a catalytic site to ORR.⁴¹ The foregoing discussion thus reveals that there is considerable uncertainty in identifying the synergistic mechanism between nitrogen and phosphorous atoms in the ORR. Thus, it is crucial to further investigate this type of catalysts from perspectives of both experiment and theory.

Herein, we constructed a controllable 3D nitrogen and phosphorous co-doped holey graphene foams (N, P-HGFs) with high specific surface area and active site density by hydrothermal and subsequent pyrolysis processes. The N,P-HGFs show a high ORR activity, i.e., a half-wave potential of 0.865 V in alkaline media, together with excellent discharge performance and long-term stability in the Zn-air battery. Importantly, it is found that the graphite N-C-P structure plays a dominant role in the N,P-HGFs for ORR from the results of both experiments and density-functional theory (DFT) calculations. This work sheds a light on the rational design and controllable synthesis of the metal-free carbon-based ORR catalysts, which further facilitates the development of the electrocatalysts in practical applications.

Experimental Section

Synthesis of catalysts

GO was synthesized by using the improved Hummers method.⁴² 1 mL of phytic acid was added to an aqueous solution of melamine (0.9 g of melamine in 20 mL of water) with stirring, and melamine-phytic acid supramolecular aggregates (MPSA) were formed by simple cooperative assembly. 10 mL of GO solution (6 mg/mL) was added to the above mixture with stirring. The precursor solution was thoroughly mixed, transferred to a hydrothermal autoclave, and hydrothermally treated at 160 °C for 6 hours. The obtained black hydrogel was repeatedly washed with water and then freeze-dried to obtain an aerogel (N,P co-doped graphene aerogel). After thoroughly grinding the 30 mg aerogel and 0.5 g of urea in an agate mortar, the mixture was placed in a quartz furnace for a high temperature heat treatment in a tube furnace. High-purity H₂ / Ar (5%: 95%) was continuously introduced, the temperature was raised to 1000 °C. After 1 h of heat preservation, the reaction was naturally cooled to room temperature to obtain a catalyst, which was recorded as N,P-HGFs-1000. The process of N-HGFs-1000 was the same as for N,P-HGFs-1000, except that no phytic acid was added. The synthesis procedure of P-HGFs-1000 is the same as that of N,P-HGFs-1000 without melamine and urea. In addition, the precursors were pyrolyzed at different

temperatures (900, 1100 °C), which was recorded as N,P-HGFs-T (T represents pyrolytic temperature). DOI: 10.1039/C9NR04696H

Characterization

The microstructure of the catalyst was observed by using a scanning electron microscope (Hitachi SU8010) and transmission electron microscopy (JEOL 2100). The composition and structure of the catalyst were also studied by a Bruker D8 diffractometer, a Raman spectrometer (HR800, JY Company) and a Physical Electronics PHI model 5700 instrument. The specific surface area and porosity of the catalyst were obtained from a Bei Shi De 3H-2000PS1. Fourier transform infrared spectra (FT-IR) were performed on a Nicolet 550II.

Electrochemical measurements

Electrochemical testing was performed by a standard three-electrode system controlled by an electrochemical workstation (CHI 760E) at room temperature, including a glassy carbon rotating disk electrode as the working electrode, mercury/mercury oxide electrode (Hg/HgO) as a reference electrode and a graphite sheet electrode as a counter electrode. The following equation was used to obtain the corresponding reversible hydrogen electrode (RHE) potential: $E(\text{RHE}) = E(\text{Hg}/\text{HgO}) + 0.896\text{V}$ in 1M KOH. First of all, 5 mg of the catalyst was added to 1000 μL of a mixed solution containing 40 μL of a Nafion solution, 340 μL of an isopropanol solution, and 620 μL of ultrapure water, and sonicated for 1.5 h to form a uniform ink. Then, 18 μL of the catalyst ink was dropped onto the glassy carbon electrode (load amount: 0.0459 mg cm⁻²). A working electrode of Pt/C was produced in a same method, and the loading of Pt/C was 0.255 mg cm⁻².

Computational details

DFT in the Kohn-Sham framework⁴³ is among the most widely used electronic-structure methods for modeling ground-state properties of molecules and condensed matter.⁴⁴ All the DFT calculations were performed employing the Dmol³ module in the Materials Studio software.⁴⁵ The exchange-correlation energy is treated at the generalized gradient approximation level within the Perdew-Burke-Ernzerhof (PBE) functional.⁴⁶ A 4×4×1 supercell was built to model both pure and co-doped graphene single sheet. The self-consistent field convergence criteria for the total energy was set to be 1.0 × 10⁻⁶ Ha during the geometry optimization for all the systems considered here. Care has been taken for choosing the *k*-point meshes in the reciprocal space and a choice of 7×7×1 is sufficient to ensure the convergence of forces and energies studied here. A vacuum region of 15 Å was employed to avoid imaginary interactions between periodic images.

Results and discussion

Materials preparation and characterization

The fabrication process of the N,P-HGFs-1000 catalyst is illustrated in Fig. 1. First, phytic acid and melamine were mixed in aqueous solution to form supramolecular aggregates (MPSA) by hydrogen bonding.⁴⁷ Then, GO aqueous solution was added into the above solution, thus GO can crosslink with

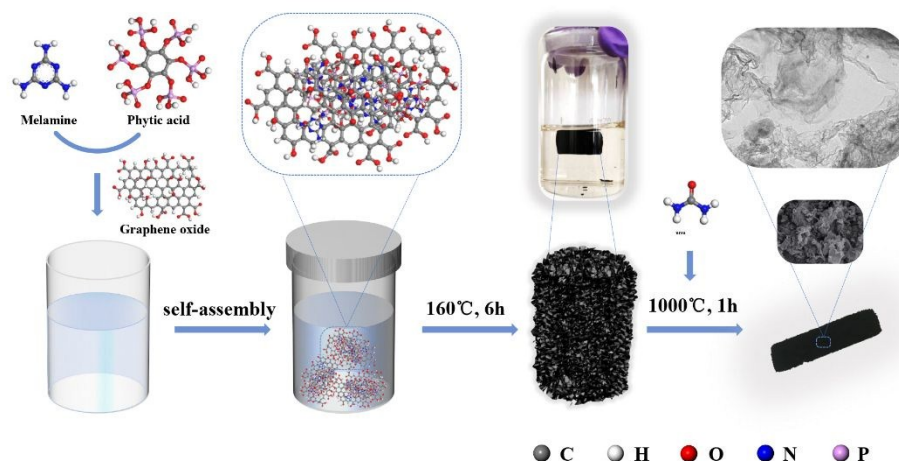


Fig. 1 Schematic illustration of the fabrication process of the N,P-HGFs-1000 catalyst.

the MPSA and form a 3D interconnected structure. The above mixture was hydrothermally reacted to help accelerate the formation of a 3D carbon network and the doping of N and P. Finally, small molecule urea was added as an additional nitrogen source and promising porogen before pyrolysis, which can result in enhanced ORR performance of the catalyst (Fig. S1).⁴ In addition, the precursors were pyrolyzed at different temperatures (900–1100 °C) to adjust the concentration of N and P contained in the synthesized catalyst. Meantime, the controlled N-HGFs-1000 without P precursor and P-HGFs-1000 catalyst without N precursor were prepared to explore the roles of N and P in the catalyst for oxygen reduction.

The SEM image (Fig. 2a) shows that N,P-HGFs-1000 exhibits graphene morphology with lamellar crimped pleats. Besides, as shown in Fig. 2b, both nitrogen and phosphorus atoms are successfully doped into the catalysts. Interestingly, compared with both N-HGFs-1000 and P-HGFs-1000, N, P-HGFs shows more folds (Fig. S2), indicating additional doping of P or N can promote the stripping stack of GO. In addition, the TEM image (Fig. 2c) of N, P-HGFs-1000 exhibits a wrinkled and transparent tulle shape, which is well consistent with the SEM image results. Besides, according to the high-resolution TEM image (Fig. 2d), one can clearly observe a typical lattice spacing, indicating a graphite (002) plane. In the N, P co-doped porous foam catalyst, the irregular structure formed by stacking and curling of graphene sheets are advantageous for increasing the charge transfer efficiency, mass transfer efficiency and effective utilization rate of the active sites, thereby promoting the ORR process.^{15, 22}

The composition and structure of the catalyst were further probed by the XRD pattern and Raman spectroscopy. It is apparent from Fig. 2a that two broad peaks around 26° and 43° were observed, which could be assigned to the (002) and (101) planes of the graphite carbon crystal structure, respectively. This result indicates that GO has been reduced to graphene. As shown in Fig. 3b, all catalysts exhibit a typical D band (~1350 cm⁻¹) corresponding to the stretching vibration of the sp² hybrid carbon atoms and a G band (~1590 cm⁻¹)

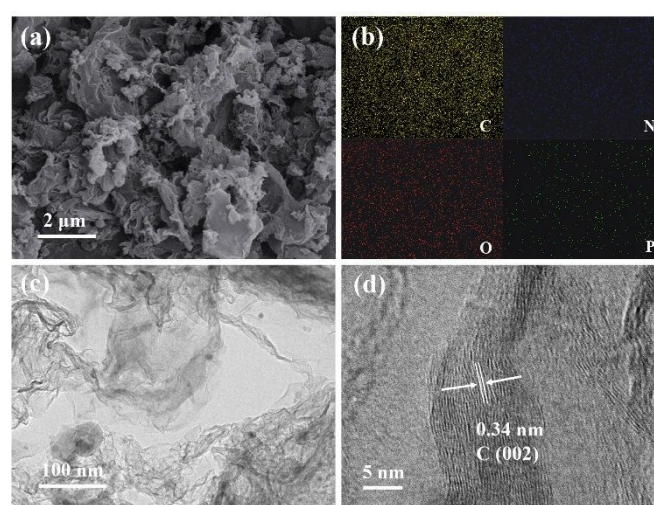


Fig. 1 (a) SEM images of the N,P-HGFs-1000 used in the EDS mapping test. (b) C, N, O and P elements mapping images. (c) TEM and (d) HRTEM images of the prepared N,P-HGFs-1000.

representing the defect of the carbon atom lattice.⁴⁸ The I_D/I_G ratio of N,P-HGFs-1000 is calculated to be 1.06, slightly larger than those of P-HGFs-1000 (1.01) and N-HGFs-1000 (1.04), indicating that more defects were formed in N,P-HGFs-1000. Also seen is the decrease of the ordered overlap of graphene sheets due to the co-doping of P and N atoms.

The specific surface area and pore structure of the catalyst have been examined by N₂ adsorption/desorption isotherm. All synthesized catalysts exhibited a type IV isotherm with a pore size distribution nearly concentrated around 4 nm (Fig. 3d). It is found that the mesoporous structure is mainly caused by the formation of the slits between graphene sheets during pyrolysis.⁴⁹ According to the calculation of BET, the specific surface areas of N,P-HGFs-1000, N-HGFs-1000 and P-HGFs-1000 catalysts are 757.73 m² g⁻¹, 693.02 m² g⁻¹ and 429.84 m² g⁻¹, respectively, which indicates that the co-doping of N and P can result in a larger specific surface area and pore volume (Table S1). It can be concluded that the resulting porous structure facilitates the exposure of more active sites, thus

preventing from their re-aggregation and improving the active site utilization and ORR mass transfer process.⁵⁰

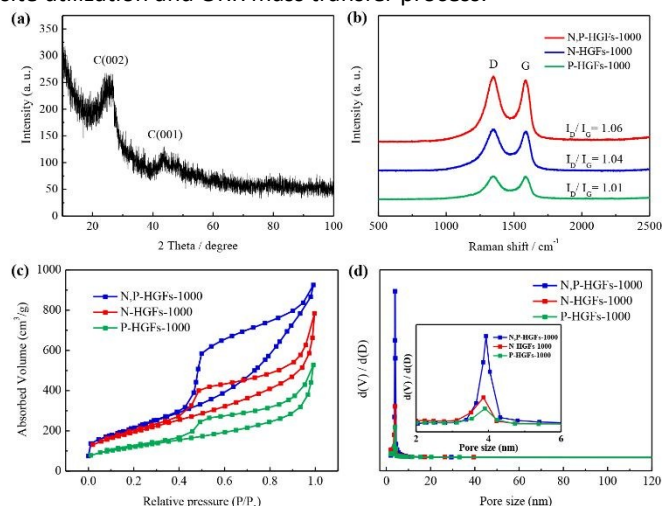


Fig. 3 (a) XRD spectra of N,P-HGFs-1000. (b) Raman spectra of P-HGFs-1000, N-HGFs-1000 and N,P-HGFs-1000. (c) N_2 adsorption-desorption isotherms of P-HGFs-1000, N-HGFs-1000 and N,P-HGFs-1000. (d) Pore size distribution curves for P-HGFs-1000, N-HGFs-1000 and N,P-HGFs-1000, the enlarged pore size distribution was shown as inset.

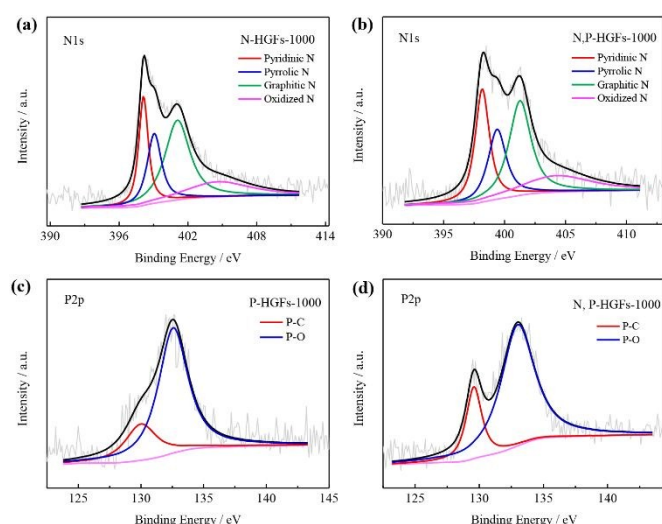


Fig. 4 High-resolution XPS spectra of N1s for (a) N-HGFs-1000 and (b) N,P-HGFs-1000. High-resolution XPS spectra of P2p for (c) P-HGFs-1000 and (d) N,P-HGFs-1000.

To investigate the effect of single doping and binary doping on the chemical composition, we have carried out the X-ray photoelectron spectroscopy (XPS) measurement for the proposed catalysts. The full XPS spectrum (Fig. S4) shows that both N and P atoms are successfully doped into the graphene carbon layer in N,P-HGFs-1000. The concentration of N and P is determined to be 2.83 and 1.22 at%, respectively (Table S2). Similarly, we conclude a successful incorporation of P and N is affirmed in both P-HGFs-1000 and N-HGFs-1000 catalysts. The high-resolution N1s spectra of N-HGFs-1000 (Fig. 4a) and N,P-HGFs-1000 (Fig. 4b) showed four peaks, which appeared at 398.2 ± 0.1 eV, 399.3 ± 0.3 eV, 401.2 ± 0.2 eV and $403 \sim 405$, respectively, corresponding to four types of nitrogen species named pyridinic N, pyrrolic N, graphitic N and oxidized N.^{13, 16} No N-P bond (~ 402.2 eV) was observed in Fig. 4d, indicating

that N and P was isolated in the graphene carbon network.⁴¹ Moreover, it is well confirmed by the infrared spectroscopy results (Fig. S5). The deconvolution of the P2p spectra of P-HGFs-1000 and N,P-HGFs-1000 catalysts showed two peaks appearing at ~ 130 eV and ~ 133 eV, which could correspond to P-C and P-O, respectively.^{51,52} As shown in Table S3, compared with N-HGFs-1000 (0.94 at%, 1.78 at%) and P-HGFs-1000 (0.27 at%), N,P-HGFs-1000 exhibits a lower content of pyridinic N (0.75 at%) and graphite N (0.96 at%) and a similar content of P-C bond (0.24 at%). In view of the synergistic effect of N and P co-doping as demonstrated by the previous works, it is expected that N,P-HGFs-1000 exhibits a higher ORR activity.^{37, 40, 48}

Electrochemical performance

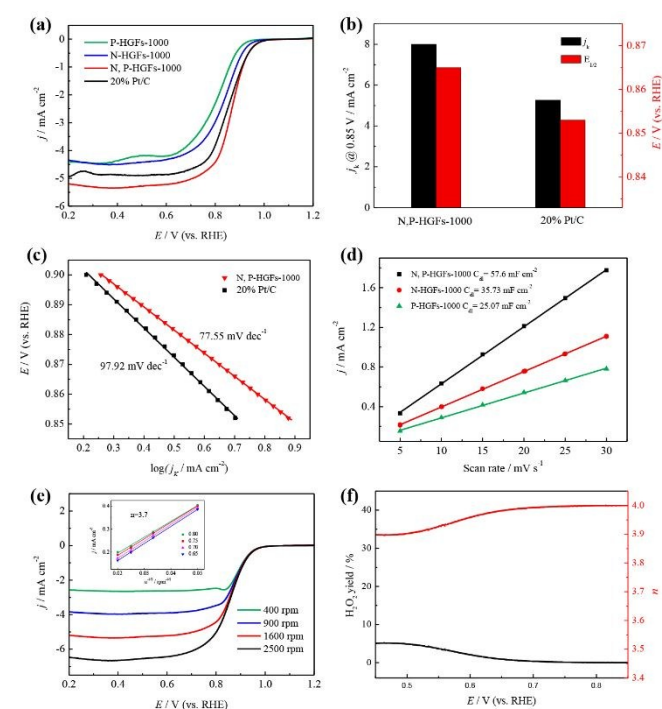


Fig. 5 (a) RDE LSV curves for P-HGFs-1000, N-HGFs-1000, N,P-HGFs-1000 and 20% Pt/C at a scan rate of 10 mV s^{-1} . (b) j_k at 0.85 V and $E_{1/2}$ for N,P-HGFs-1000 and Pt/C. (c) Tafel plots of N,P-HGFs-1000 and 20% Pt/C. (d) Plots of current densities (at 1.1 V vs. RHE) as different scanning speed. (e) LSV curves of N,P-HGFs-1000 at a series of rotation speed from 400 rpm to 1600 rpm (inset: the fitted K-L plots at different potentials). (f) H_2O_2 yields and the n for N,P-HGFs-1000.

The ORR performance of N,P-HGFs-1000 is assessed by the linear sweep voltammetry (LSV) at 1600 rpm . The N,P-HGFs-1000 catalyst exhibits the highest ORR performance with respect to onset potential (E_{onset}) of 0.983 V , which is 1.06 times and 1.03 times of that for P-HGFs-1000 and N-HGFs-1000 (Fig. 5a). Moreover N,P-HGFs-1000 presents a half-wave potential ($E_{1/2}$) of 0.865 V , even better than Pt/C (0.853 V), indicating that the co-doping of P and N is in favor of stimulating the ORR performance. Furthermore, the kinetic current density (j_k) is obtained via equation (1) (Fig. 4e). The N,P-HGFs-1000 catalyst exhibits an excellent j_k of 7.99 mA cm^{-2} at 0.85 V , about 1.5 times as that of Pt/C (5.27 mA cm^{-2}) (Fig. 5b). As presented in Fig. 5c, the Tafel slope, which is regarded as a crucial indicator of the dynamic rate on ORR process, is

77.55 mV dec⁻¹ and 97.92 mV dec⁻¹ for N,P-HGFs-1000 and Pt/C, respectively, demonstrating a remarkable kinetic process of ORR on N,P-HGFs-1000. Beyond that, the double layer capacitance (C_{dl}) of N, P-HGFs-1000 is 57.6 mF cm⁻², which is superior to P-HGFs-1000 (25.07 mF cm⁻²) and N-HGFs-1000 (35.73 mF cm⁻²), proving the superior electrochemical surface area of N, P-HGFs-1000, which is well aligned with the performance results of Fig. 5a.

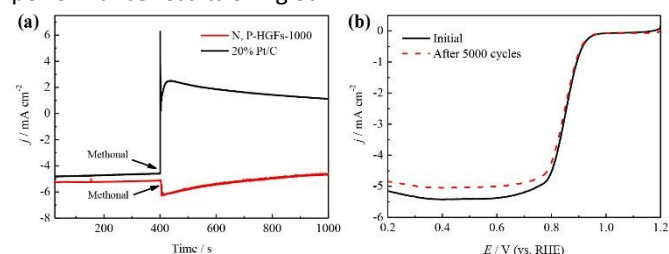


Fig. 6 (a) Current-time (*i-t*) chronoamperometric response of N,P-HGFs-1000 and Pt/C electrodes by adding 6.78 mL methanol into the 50 mL solution after about 400 s at 0.65 V (vs RHE) with 1600 rpm. (b) LSV curves before and after 5000 cycles of N,P-HGFs-1000.

By computing the number of electrons transferred (n) and the j_k , which are performed by a series of polarization curves with different rotation rates, the effectiveness of a catalyst can be estimated using the K-L equation. The value of n is approximately 3.7, suggesting a satisfactory four-electron ORR pathway for N,P-HGFs-1000 (Fig. 5e). Using rotating ring-disk electrode (RRDE) measurement, it is easy to calculate n and to monitor the evolution for the yield of peroxide (H_2O_2) during the ORR process. As presented in Fig. 5f, the H_2O_2 yield of N,P-HGFs-1000 catalyst is below 5% and the n ranges from 3.9 to 4.0 at the potential range of 0.45 V–0.85 V, which is similar to the result from K-L equation.

The chronoamperometry (*i-t*) was used to obtain the effect of methanol on the catalytic activity of N,P-HGFs-1000. When methanol was injected, the current density on the N,P-HGFs-1000 electrode changed slightly, indicating excellent tolerance of N,P-HGFs-1000 to methanol crossover. On the contrary, an instantaneous current jump occurred at the Pt/C electrode, after which the methanol oxidation reaction completely inhibited the progress of the ORR. The N, P-HGFs-1000 have more effective resistance to methanol permeation than Pt/C. The ORR stability of the catalyst was displayed by cyclic voltammetry in an O_2 -saturated electrolyte at 50 mV s⁻¹ between 0.6 and 1.0 V using an accelerated durability test protocol (ADT). After 5000 cycles under the same condition, the performance of N,P-HGFs-1000 was only attenuated by 5 mV (Fig. 6b). However, due to surface oxidation and particle dissolution/aggregation, the ORR performance of Pt/C was significantly reduced (30 mV) (Fig. S10). These results indicate a significant advantage of non-metal N,P-HGFs-1000.

Inspired by the efficient ORR activity of N,P-HGFs-1000 in an alkaline medium, the N,P-HGFs-1000 was used as an electrocatalyst in an air cathode to construct a primary Zn-air battery to explore the potential practical application of the obtained catalyst. Schematic illustration of the Zn-air batteries are presented in Fig. 7a. For reference, an additional Zn-air

battery was prepared with the same loading of 20% Pt/C catalyst as the cathode. The battery using N,P-HGFs-1000 catalyst (1.452V, 103 mW cm⁻²) exhibits a similar open circuit voltage and peak power density as the battery containing Pt/C (1.456V, 102 mW cm⁻²) (Fig. 7b.), which is comparable with most of the catalysts have been reported (Table S5). In addition, the specific capacity of Zn-air batteries with N,P-HGFs-1000 normalized to the mass of the consumed Zn at 10

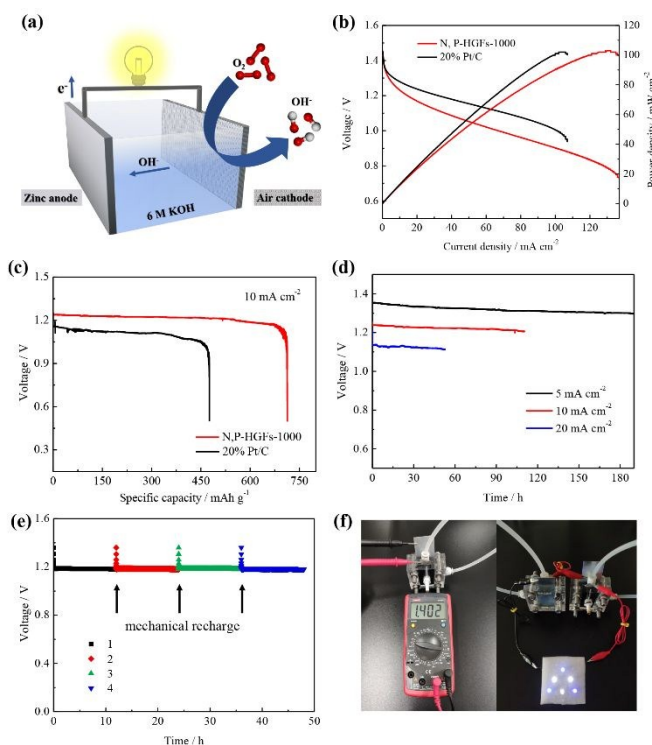


Fig. 7 (a) Schematic illustration of the primary Zn-air battery. (b) Polarization and power density curves of the Zn-air batteries using 20% Pt/C and N,P-HGFs-1000 as ORR catalyst at a scan rate of 5 mV s⁻¹. (c) Specific capacities of the Zn-air batteries using 20% Pt/C and N,P-HGFs-1000 as ORR catalyst. (d) Discharge curves of the primary Zn-air batteries using N,P-HGFs-1000 as ORR catalyst. (e) Long-time durability of the N,P-HGFs-1000 based Zn-air battery at 10 mA cm⁻² with mechanical recharge. (f) A photograph of the Zn-air battery with N,P-HGFs-1000 measured with a multimeter; and a photograph of lighted LEDs (2.2V) powered by two N,P-HGFs-1000 based Zn-air batteries using the connected in series.

mA cm⁻² is 713 mAh·g_{Zn}⁻¹ (corresponding to the gravimetric energy density of ~864 Wh g_{Zn}⁻¹), which is significantly higher than Pt/C based batteries (476 mAh·g_{Zn}⁻¹, 527 Wh g_{Zn}⁻¹) (Fig. 7c). Subsequently, the constant current discharge curves (~190h, ~110h and ~53h) at 5, 10 and 20 mA cm⁻² are presented in Fig. 7d, indicating a stable discharge capacity of the battery. The Zn-air battery using N,P-HGFs-1000 catalyst was continuously discharged for 48 hours at a current density of 10 mA cm⁻², and the voltage remained substantially unchanged during the four mechanical recharge cycles (Fig. 7e), which indicates a long-term durability of Zn-air batteries with N,P-HGFs-1000. As shown in Fig. 7f, the N,P-HGFs-1000 based Zn-air battery shows a open circuit voltage of 1.402V measured with a multimeter. Moreover, two batteries can be series-connected for illuminating LEDs of different colors (5 mm, 2.2 V). And, the fan (1.0 V) can be powered by only one

piece of Zn-air battery (Video S1). These results confirm that N,P-HGFs-1000 catalysts exhibit great potential for Zn-air batteries in green energy storage and conversion.

ORR mechanism of the N,P-HGFs catalyst

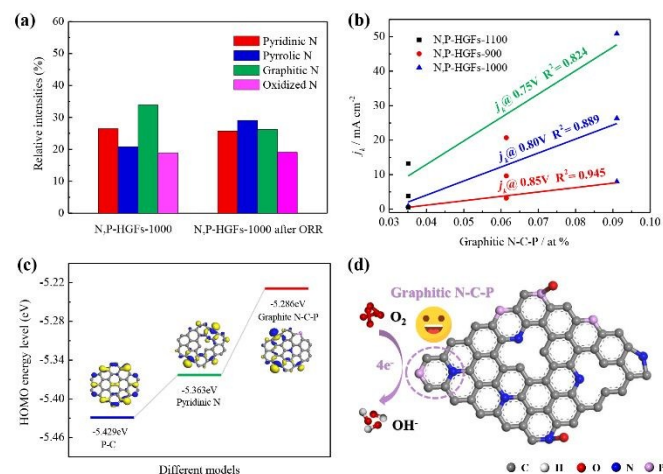


Fig. 8 (a) The content of different types of nitrogen for N,P-HGFs-1000 catalyst and N,P-HGFs-1000 catalyst after ORR. (b) Correlation between j_k at 0.85, 0.8, and 0.75 V and the graphitic N-C-P species concentrations. (c) HOMO energy levels on graphitic N-C-P, pyridine N and P-C models. (d) The mechanism scheme of graphitic N-C-P-dependent ORR.

The source of ORR performance of the N,P-HGFs catalyst was further analyzed by ex situ post-ORR XPS. As shown in Fig. 8a, the concentration of the pyridinic N species did not change substantially after the start of the polarization scan, which was inconsistent with previous work, demonstrating that pyridinic N is not the major active site in N,P-HGF-1000.^{29,53} To further verify the above hypothesis, the relationship between the j_k of the catalyst and the concentration of different dopants was further studied. As shown in Fig. S12a, the correlation between j_k and the pyridine N concentration could not be observed. Similarly, there was no significant correlation between ORR activity and the content of pyridine N-C-P species (Fig. S12b), individual graphite N species (Fig. S12c) or P-C species (Fig. S12d). Conversely, a strong linear correlation was observed at 0.85 V between the ORR activity of the N,P-HGFs catalyst and graphitic N-C-P species (Fig. 8b) with a correlation parameter R^2 of 0.945, which indicate the ORR activity at high potential was dominantly determined by the concentration of graphitic N-C-P. Besides, the calculated turnover frequency of graphitic N-C-P is 2.5 electrons per site per s at 0.85 V, which is much higher than that of pyridinic N or P-C (Fig. S13), thus demonstrating the dominated role of the graphitic N-C-P structure.

Furthermore, the catalytic mechanism of the graphitic N-C-P structure was further revealed by DFT calculations. The electron cloud distribution of the highest occupied molecular orbital (HOMO) in different catalyst models is shown in Fig. 8c. The highest HOMO energy (-5.286 eV) of the graphite N-C-P structure not only represents the lowest energy gap between the catalyst and the lowest unoccupied molecular orbital (LUMO) of O_2 , but also the maximum overlap of the molecular orbitals of the two boundaries.^{30, 37} As a result, the electron

transfer to the O_2 molecule is more readily achieved in the N,P-HGFs-1000 catalyst containing more graphitic N-C-P structure than that in the catalysts containing more other structures; thus ORR is accelerated within N,P-HGFs-1000. On the other hand, when nitrogen atoms are doped into the carbon skeleton, they tend to induce an increased charge density of the carbon atom adjacent to nitrogen, resulting in positively charged carbon atoms.⁵⁴ It can be concluded that the positively charged carbon atoms are active sites, which represent electron donors to the adsorbed O_2 .²⁹ Interestingly, the charge density of the adjacent C is further increased (Fig. S14) when the doped P atoms are bonded to C adjacent to the graphite N, yielding an increase in catalytic activity.

Conclusions

In summary, we have designed a controllable 3D nitrogen and phosphorus co-doped graphene catalyst by hydrothermal and subsequent pyrolysis processes. The obtained N,P-HGFs showed high ORR activity (i.e., half wave potential of 0.865 V), long-term stability and excellent methanol resistance, which is superior to 20% Pt/C. In addition, the Zn-air battery using N,P-HGFs-1000 as the ORR catalyst has shown improved discharge performance and excellent durability, indicating that our catalyst represents a promising candidate for the energy transfer and storage. Importantly, the experimental results and DFT calculations demonstrate that the C atom with positive charge in the graphite N-C-P structure is the dominated ORR active site in N,P-HGF catalysts. This work reveals a rational design and controllable synthesis of heteroatom-doped porous carbon-based ORR catalysts, paving a new way of the widespread application of electrocatalysts.

Conflicts of interest

There are no conflicts to declare.

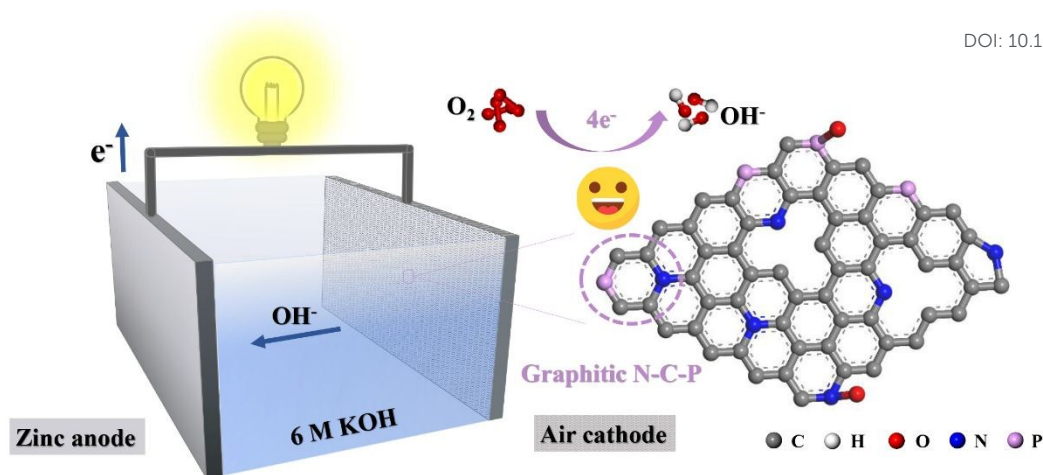
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The C atom with positive charge in the graphite N-C-P structure is the dominated ORR active site in N,P-HGFs.